A sharp capacity boundary for budget-feasible EF1 chore allocations

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Abstract

Consider complete chore allocations with common processing times in a fixed interval $[a, b]$ and a common time budget B, where $0 < a \le b \le B$. For any fixed number $n \ge 2$ of agents, the best uniform EF1 factor is 1 if $\lfloor B/a \rfloor = \lfloor B/b \rfloor$, and is unbounded otherwise. The positive case follows from least-cost round robin. In the negative case, a packing argument forces every feasible allocation to have the same unfair shape, even with strictly positive identical disutilities. Thus arbitrarily small differences in processing times can destroy every finite guarantee.

1 Model and statement

There are n agents and a finite set C of indivisible chores. Chore c has a common processing time $t_c \in [a, b]$, and every agent has budget B. Agent i has additive nonnegative disutility $d_i(S) = \sum_{c \in S} d_i(c)$. An allocation $A = (A_1, \dots, A_n)$ is a partition of C, and is feasible if $\sum_{c \in A_i} t_c \le B$ for all i. We consider only instances admitting a feasible allocation.

For $\alpha \ge 1$, call A α-EF1 if for every $i \ne j$ either $A_i = \varnothing$, or some $c \in A_i$ satisfies

$$d_i(A_i \setminus \{c\}) \le \alpha d_i(A_j).$$

The deleted chore belongs to the potentially envious agent. In this common-time, common-budget setting every feasible bundle fits every agent, so this is also the budget-aware full-bundle comparison proposed in catalogue entry D63-448. Let $\alpha(A)$ be the infimum of the admissible factors, with $\inf \varnothing = +\infty$, and define the uniform guarantee

$$G_n(a, b; B) = \sup_{I} \min_{A \text{ feasible}} \alpha(A),$$

where I ranges over the instances just specified. The bounds a, b need not both occur as processing times in an instance.

Theorem 1. *For every $n \ge 2$ and $0 < a \le b \le B$,*

$$G_n(a, b; B) = \begin{cases} 1, & \lfloor B/a \rfloor = \lfloor B/b \rfloor, \\ +\infty, & \lfloor B/a \rfloor > \lfloor B/b \rfloor. \end{cases}$$

In the first case, least-cost round robin constructs a feasible EF1 allocation. In the second, the guarantee remains unbounded for identical strictly positive disutilities and only two processing times.

Elkind's problem in the Dagstuhl report [1, Section 4.1] concerns fairness with agent-dependent chore times and asks for suitable envy notions. The displayed approximation criterion is the catalogue's formulation, not a definition adopted by that report. A prior catalogue attempt [3] already gives an unbounded-factor example and proves the equal-duration round-robin guarantee. Theorem 1 determines the exact boundary for every fixed interval of common times. It does not address the broader heterogeneous-time axiomatization agenda. The budget model of Elkind, Igarashi, and Teh [2, Section 2.1] permits an auxiliary housekeeper; here every chore must be allocated to the n agents.

2 The capacity test is sufficient

Write $k = \lfloor B/b \rfloor$. If $\lfloor B/a \rfloor = k$, every feasible bundle has at most k chores. Hence a feasible instance with m chores satisfies $m \leq nk$.

Run round robin in a fixed agent order, each agent choosing a remaining chore of minimum disutility on her turn. Bundle sizes differ by at most one, so each agent receives at most $\lceil m/n \rceil \leq k$ chores. Every resulting workload is therefore at most $kb \leq B$.

For completeness, fix agents $i \neq j$ with $A_i \neq \varnothing$ and remove the last chore chosen by i. If i precedes j in the round, compare each retained rth choice of i with the rth choice of j. If i follows j, compare it with the $(r+1)$st choice of j. In either case the comparison choice exists, was still available when i chose, and differs for different retained chores. Minimum-disutility selection and nonnegativity give

$$d_i(A_i \setminus \{c\}) \leq d_i(A_j).$$

Thus the allocation is EF1. Since factors are constrained to be at least one, the guarantee is exactly one. With rational inputs, the construction takes a polynomial number of exact comparisons.

3 A capacity gap is sufficient for unboundedness

Suppose $\lfloor B/a \rfloor > k = \lfloor B/b \rfloor$. Then $k \geq 1$ and

$$a \leq s := \frac{B}{k+1} < b.$$

Fix $0 < \varepsilon \leq 1$. Use $(n-1)k$ cheap chores, each with time b and disutility ε, and $k+1$ expensive chores, each with time s and disutility 1. All agents have these same disutilities. The instance is feasible: one agent receives all expensive chores, and every other agent receives k cheap chores.

There are $nk+1$ chores, so in every complete allocation some agent receives at least $k+1$ chores. Every chore takes at least s, and $(k+1)s = B$. A feasible bundle of that size consequently has exactly $k+1$ chores and contains no cheap chore, since $b > s$. It therefore contains all expensive chores. Each other bundle consists only of cheap chores and contains at most k of them. Completeness forces each of these $n-1$ bundles to contain exactly k cheap chores.

After any deletion, the expensive-bundle agent has disutility k and compares it with $k\varepsilon$. Thus every feasible allocation requires $\alpha \geq 1/\varepsilon$. This factor also suffices: the expensive-bundle inequalities hold with equality, and each cheap-bundle agent has residual $(k-1)\varepsilon$, no greater than any comparison disutility after multiplication by $1/\varepsilon$. Hence every feasible allocation satisfies

$$\alpha(A) = 1/\varepsilon.$$

Letting ε decrease to zero proves the second case of Theorem 1. All data in the construction are rational whenever a, b, B, ε are rational.

Corollary 2. *For every fixed $n \geq 2$ and every $R > 1$, no finite factor is guaranteed over feasible common-time instances with a nonempty chore set, equal budgets, and $\max_c t_c / \min_c t_c \leq R$, even with identical strictly positive disutilities. At $R = 1$, factor one is always attainable.*

Proof. Take $B = 2$, $a = 1$, and $b = 1 + \eta$ with $0 < \eta < \min\{R - 1, 1\}$. The two capacities are two and one, so the negative case applies. When $R = 1$, all processing times are equal and the capacities coincide. □

Use of AI tools

The author used research AI agents that the author built and configured (GPT-6 Astra max, Claude Opus 5.5 max and GPT-6 Astra Pro) during the proof work, for proof search and construction, proof checking, translation, Python computation and similar tasks. The final manuscript was written, and the proofs were checked, by the human author. All responsibility for this work rests with the author personally.

References

[1] E. Elkind. Fair Division of Chores with Budget Constraints. Section 4.1 in E. Markakis, R. Mehta, and Y. Zick (eds.), *Fair Division: Algorithms, Solution Concepts, and Applications*, Dagstuhl Reports 14(9), pp. 145–166, 2025. https://doi.org/10.4230/DagRep.14.9.145.

[2] E. Elkind, A. Igarashi, and N. Teh. *Fair Division of Chores with Budget Constraints.* SAGT 2024, LNCS 15156, pp. 55–71, 2024. https://doi.org/10.1007/978-3-031-71033-9_4. Full text: https://arxiv.org/html/2410.23979v1.

[3] GPT 6 Astra Ultra. *An unbounded-factor obstruction and an equal-duration EF1 theorem.* Prior research attempt for D63-448, Problem Hunting with LLMs. https://github.com/mehmetmars7/Erdosproblems-llm-hunter/blob/00ced55ba1223fb48603e1f15028d753997ee515/attacks/open_problems/economics/gpt_6_astra_ultra/D63-448.tex.